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RESEARCH ARTICLE

Football Player Injury Full-Cycle Management and Monitoring System Based on Blockchain and Machine Learning Algorithm

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Abstract
Football injuries are the most common factor affecting a football player's performance, and the last thing a football player wants. To understand the causes of football players' injuries and how to recover sports injuries most efficiently, the football players' injuries were managed and monitored throughout the whole cycle. However, the traditional football player injury cycle management and monitoring system are not only insecure in data storage, but more importantly, it lacks intelligent analysis of the collected data. With the continuous development of blockchain and machine learning technologies, blockchain technology is used to collect, store, clean, mine and visualize the full-cycle data of football players' injuries, and machine learning is used to provide intelligent solutions for football players' injury recovery. This paper compared the football player's injury full-cycle management and monitoring system based on blockchain and machine learning algorithm with the traditional football player's injury management and monitoring system. The experimental results showed that the average self-processing capacity of the football player injury MMS based on blockchain and ML algorithm was 70%, while the average self-processing capacity of the traditional football player injury management and monitoring system was 50%. Therefore, the application of blockchain and machine learning algorithm in the football player's injury full-cycle management and monitoring system can effectively improve the system's self-processing ability.

Keywords Full-cycle management and supervision system · Football player injury · Machine learning algorithm · Blockchain technology · Support vector machine

1 Introduction

Under the background of globalization, people pay more and more attention to the development of sports. Football has gradually developed into the first sport in the world due to its high degree of confrontation, complex field tactics and large number of participants. In the fierce football game, football players are injured very frequently, which seriously affects their play on the field. The management and supervision of the injury process of football players would help to analyze the causes of injury and the therapeutic effects of various injury treatments. At the same time, under the trend of extensive application of deep learning algorithm, the establishment of interactive intelligent medical prediction and evaluation model based on deep learning model provides a corresponding reference for the prediction and evaluation of intelligent medical in the later stage [1]. Therefore, it is very important to build a football player injury full-cycle management and supervision system. However, the traditional football player injury full-cycle management and monitoring system is mainly for the team medical personnel to record the injury of football players. The management and supervision system relies on manual records, and cannot intelligently analyze the recorded injury data, which makes the traditional football player injury full-cycle management

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and supervision system of little practical significance. With the continuous development of blockchain technology, it is possible to realize the permanent preservation of data, and use blockchain technology to collect and save the data of football players' injury cycle. This paper analyzed the whole cycle monitoring data of football players' injuries, and used machine learning to transform the injury data into decision analysis. By generating an intelligent treatment scheme for football players' injuries, the automatic processing ability of football players' injury full-cycle management and monitoring system can be improved, so this paper has research significance. In the fierce football match, football players are often injured, which seriously affects their performance on the field. The traditional management and monitoring system of injury and disease cycle of football players is not only insecure in data storage, but also lacks intelligent analysis of collected data. The research problem of this paper is the cause of the injury of football players and how to recover the sports injury most effectively.

2 Related Work

Football is a very fierce sport, and it is inevitable for football players to have sports injuries during training or competition. The management and supervision of football players' injuries throughout the whole cycle is conducive to the recovery of sports injuries. Guo et al. compared the specific movements of football players before and after the fatigue exercise, and compared the characteristics of technical movements of football players using three-dimensional infrared rays. The results showed that the body of football players trembled after fatigue, which brought great pressure to the knee joint [2]. Sun and Feng carried out management and supervision on the injury cycle of high school football players. He analyzed the types of football sports injuries of senior high school students, and helped senior high school football players recover quickly according to the past experience of injury recovery [3]. Tee et al. pointed out that the fundamental of preventing sports injuries depended on the supervision and analysis of the whole process of sports, and the management and supervision of athletes' sports injuries can quickly find out the causes of sports injuries [4]. Kerr used the Internet to monitor the sports injuries of American high school football for 10 years, and analyzed the main causes of sports injuries of football players. This could provide a large amount of data for the treatment of football sports injuries and effectively improves the recovery effect of football sports injuries [5]. It is helpful to analyze the cause of injury of football players and improve the effect of injury recovery by full-cycle management and supervision of injury of football players. However, there is a lack of intelligent technology to intelligently collect and process the injury information of football players.

With the continuous development of intelligent technology, blockchain and machine learning algorithms can intelligently collect, analyze and visualize data. Many scholars have applied blockchain and machine learning algorithms to the football player's injury full-cycle management and monitoring system. Zhang and Ai used blockchain technology to record the specific details of the athletes' injuries and the time of injury, and used machine learning algorithms to intelligently analyze the characteristics of the athletes' injuries, so as to recommend targeted sports injury rehabilitation programs for athletes [6]. Wang showed that football injuries were too frequent. Using blockchain and machine learning algorithms to build an intelligent football injury full-cycle management and monitoring system, it can effectively conduct intelligent analysis of football injury [7]. Tian et al. proposed a particle swarm optimization (VSMPSO) based on variable proxy model to solve this challenge, and extended it to solve the 200-dimensional problem. Specifically, the single alternative model constructed by simple random sampling is used to explore different promising areas in different iterations. The results show that the proposed VSMPSO can achieve high-quality solutions and computational efficiency for high-dimensional problems [8]. Meng et al. proposed a network public opinion crisis rating method based on multi-level indicator system to objectively evaluate the impact of events. First, he explained the transmission mechanism of online public opinion from the perspective of information ecology. According to this mechanism, he selected some evaluation indicators through correlation analysis and principal component analysis. Then, he created a text emotion classification model through deep learning training to achieve accurate quantification of emotion indicators in the index system [9]. Han et al. discussed a total of 131 articles for literature measurement, and obtained the visualization map and main research topics in the field of PCSCM (prefabricated construction supply chain management). The research results show that the intelligence and informatization of supply chain is the focus of current research. Finally, he pointed out the research directions that should be paid attention to in the field of PCSCM [10]. The research of Van shows that a large number of athletes' sports injury data can be obtained and analyzed by machine learning, which can accurately predict sports injuries [11]. The application of blockchain and machine learning algorithm in the football player injury full-cycle management and monitoring system can improve the data analysis ability of football player injury. However, there is a lack of comparison with the traditional football player injury management and monitoring system.

Football sports injuries refer to all kinds of injuries that occur to athletes during football training or competition. Due

to the large degree of football activities, the injured parts and degrees are different [12, 13]. This paper used blockchain and machine learning algorithm to analyze the data of football player's injury full-cycle MMS, and compares it with the traditional football player's injury full-cycle MMS. The results showed that the football player injury full-cycle MMS based on blockchain and ML algorithm has higher data security.

3 Methods of Full-Cycle Management and Supervision of Sports Injury

Sports injuries are very common, and the parts of injuries are directly related to the sports training that athletes themselves receive. For example, in football, the injured parts of athletes are the knee joint, ankle joint and hip joint. This is because football requires athletes to run with high intensity, as well as football players' insufficient training level, inaccurate actions and lack of awareness of protection [14, 15]. To a large extent, the arrangement of sports injury rehabilitation for football players only depends on the suggestions of team doctors and the way of rehabilitation training, but the rehabilitation effect is not good. It is necessary to manage and supervise the whole injury cycle of football players and promote the recovery of football from different perspectives. The functional model of football injury full-cycle management and supervision is shown in Fig. 1, which is adopted from reference [16].

In Fig. 1, the objects of football sports injury full-cycle management and supervision are divided into three categories, namely football team doctors, football coaches and football players. Through the collection of basic information, player training plan, nutrition data, physical fitness data and recovery data of football players, the full-cycle management has monitored the injury of football players.

3.1 Blockchain Technology

Blockchain is a digital technology that supports digital currency system. It has such attributes as decentralization, distrust, tamper proof, collaborative maintenance, timing data, etc. The distributed football player injury and disease full-cycle management monitoring system realizes decentralized point-to-point data transmission without node trust. This effectively solves the problems of high cost, low efficiency and data security of centralized data transmission. Blockchain is a distributed database system in essence, which is characterized by data tampering and data decentralization. The structural model of the blockchain is shown in Fig. 2, which is adopted from reference [17].

Figure 2 describes the application model of blockchain in the football player's injury full-cycle management and monitoring system. The overall structure is divided into four layers. The higher the hierarchy, the more user oriented. The application layer is used by managers in the football player injury full-cycle management and monitoring system, which mainly realizes the acquisition and storage of football player injury full-cycle management and monitoring data through blockchain technology.

The technical features of blockchain application in football injury full-cycle management and monitoring system include as follows:

Decentralization It means that the entire network has no centralized hardware or management organization. The rights and obligations of any node are equal, and the damage

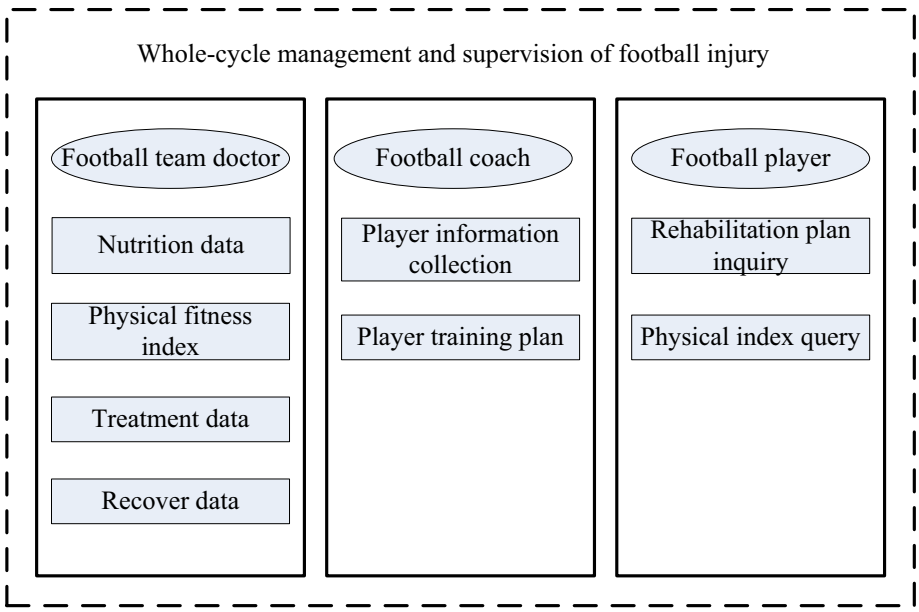


Fig. 1 Functional model of football injury full-cycle management and supervision

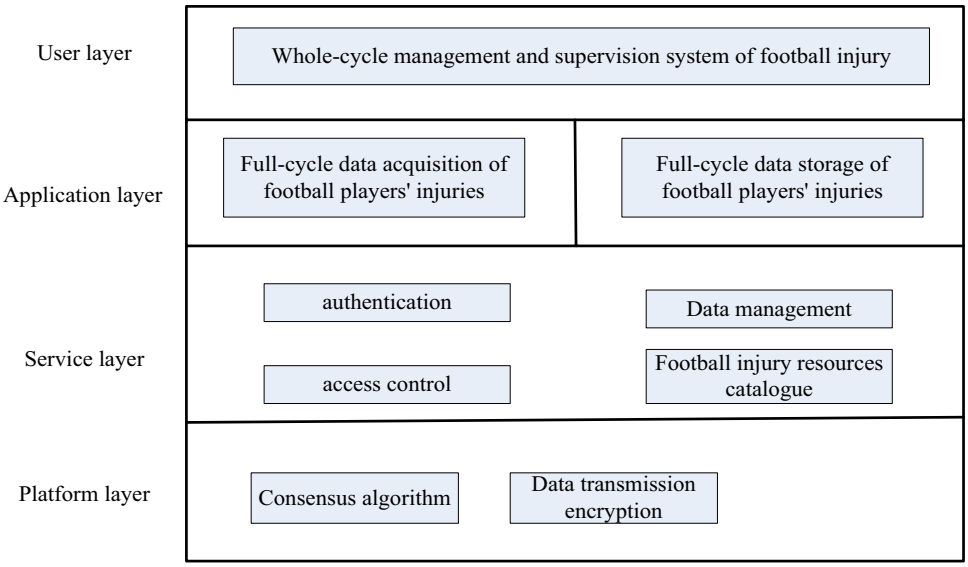


Fig. 2 Blockchain structure model

or loss of any node would not affect the operation of the whole system [18].

Deconfidence It means that nodes in the whole system can communicate data without mutual trust. All operations of the system are open and transparent, and all data contained therein are also transparent.

Maintenance collectivization All participating nodes maintain the data in the system together, and each node can perform maintenance operations on the data.

In recent years, the blockchain technology has been developing continuously, and the scenarios of blockchain applications have become more and more extensive, which are widely used in medical, education and government departments. Due to the extensive sources of data in the football player injury lifecycle management and monitoring system, multiple storage nodes are required. To ensure the consistency of football player injury data, the stored data need to be decentralized. Therefore, the blockchain can be well applied to the football player's injury full-cycle management and monitoring system to effectively obtain and store complex data in the sports injury full-cycle management and monitoring system.

3.2 Machine Learning Algorithm

Machine learning is an interdisciplinary field, including the research on probability theory, statistics and intelligent algorithms. It can intelligently analyze complex information data, which is the basis of artificial intelligence. It can well analyze football players' injury full-cycle management and monitoring data. There are many research directions of machine learning, including artificial neural networks, naive Bayesian classification and support vector machines.

In the football player injury full-cycle management and monitoring system, the data collected by the system are numerous and disorderly. It is difficult to find out the information related to the treatment of sports injury and the full-cycle management and supervision information of a sports injury from the football player's injury full-cycle management and supervision system [19, 20]. Support vector machine (SVM) is a kind of generalized linear classifier that classifies data into two classes according to supervised learning. The decision boundary is the maximum margin hyperplane of learning samples, and the data are classified by kernel function. The structural model of support vector machine is shown in Fig. 3, which is adopted from reference [21].

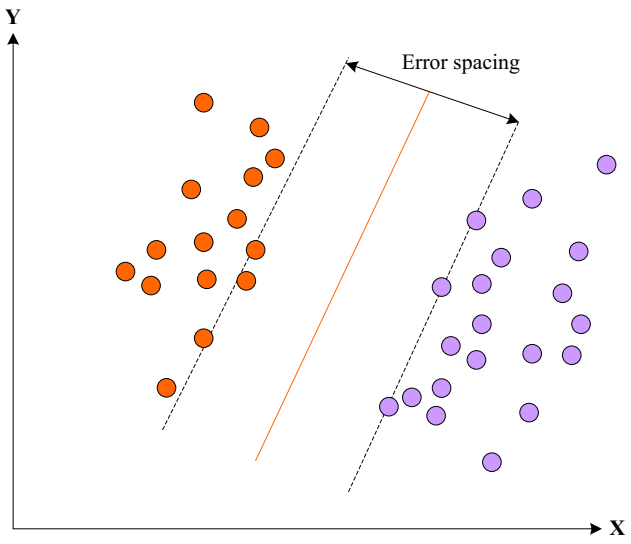


Fig. 3 Structural model of support vector machine

Figure 3 describes the structural model of support vector machine. In the football player injury full-cycle management and monitoring system, there are different data characteristics between different data. The hyperplane can be used to separate the data according to the data characteristics, such as the treatment effect of football players' knee joint injuries after different drug treatments.

The process of using support vector machine to analyze football players' injury full-cycle management and monitoring system is as follows: n training samples are selected from football players' injury full-cycle management and monitoring system, and the training samples are expressed as:

$$(X, Y) = \left\{ \left(x_1, y_1\right), \left(x_2, y_2\right), \ldots, \left(x_n, y_n\right) \right\}. \tag{1}$$

In Formula (1), x_n represents the input data of the nth sample data, such as the information of the injured part of a football player. y_n represents the output data of the nth sample data, such as an effective treatment scheme for football sports injuries.

The regression function of support vector machine is

$$g(x) = cu(x) + d. \tag{2}$$

In Formula (2), $g(x)$ represents regression function, and $u(x)$ represents mapping function; c and d are regression parameters of regression function.

The process of vector analysis of football player's injury full-cycle management and supervision system can be transformed into the process of solving regression parameters c and d. The process of solving parameters c and d can be expressed as

$$\begin{cases} \begin{cases} y_i - cu\left(x_i\right) - d \le b + a_i, \\ -y_i + cu\left(x_i\right) + d \le b + a_i^{'} \\ a_i, a_i^{'} \ge 0 \end{cases} \\ \min \dfrac{1}{2}\|c\|^2 + E \sum\limits_{i=1}^{n} \left(a_i + a_i^{'}\right). \end{cases} \tag{3}$$

In Formula (3), the value range of i is $\{1, 2, 3, \ldots, n\}$, and E represents the penalty factor; variables a_i and $a_i^{'}$ are relaxation variables, and b represents the error distance of the classification of football players' injury full-cycle data.

The classification characteristics of football players' injury full-cycle data can be solved through the error distance

$$R(g(x), y, b) = \begin{cases} |y - g(x)| - b, |y - g(x)| > b \\ 0, \text{other} \end{cases}. \tag{4}$$

In Formula (4), $R()$ represents the classification characteristic function.

By jointly solving the equations in Formula (3), the solution of regression parameters c and d can be obtained, and the regression function is expressed as

$$g(x) = \sum_{i=1}^{n} c_1 Z(x) + d_1. \tag{5}$$

In Formula (5), c_1 and d_1 are the solutions of regression parameters c and d, respectively, and $Z(x)$ is the kernel function.

3.3 Football Player Injury Full-Cycle Management and Supervision System

There are more and more football sports injuries. However, the existing football players' injury full-cycle management and monitoring data are not comprehensive and intelligent enough, which leads to the lack of stable football sports injury rehabilitation treatment [15, 16]. The equipment such as the Internet of Things can be used to obtain the full-cycle data of football players' injuries, and use the blockchain technology to decentralize and store the acquired data. Finally, the data of past football players' injuries are comprehensively analyzed to simplify the operation of management and supervision personnel. The model of football player injury full-cycle management and monitoring system is shown in Fig. 4, which is adopted from references [22, 23].

Figure 4 describes the football player injury full-cycle management and monitoring system model using blockchain and machine learning algorithm. First of all, it obtained the full-cycle management and monitoring data of football player injuries. With the help of blockchain technology, it stored and converted the data, so as to build a football player injury data management platform. After that, the machine learning algorithm is used to intelligently analyze the injury data of football players and provide decision-making analysis for the injury treatment of football players.

The football player's injury full-cycle management and supervision system carries out intelligent management on the player's injury recovery under the information condition. The significance of its construction makes the football sports management and supervision gradually transform the management of football players from the original manual to intelligent. In this system, the injury, treatment and rehabilitation of football players are comprehensively considered, which makes the rehabilitation process of football players become comprehensive and scientific, so as to better promote the training and recovery of football players after injury [24, 25].

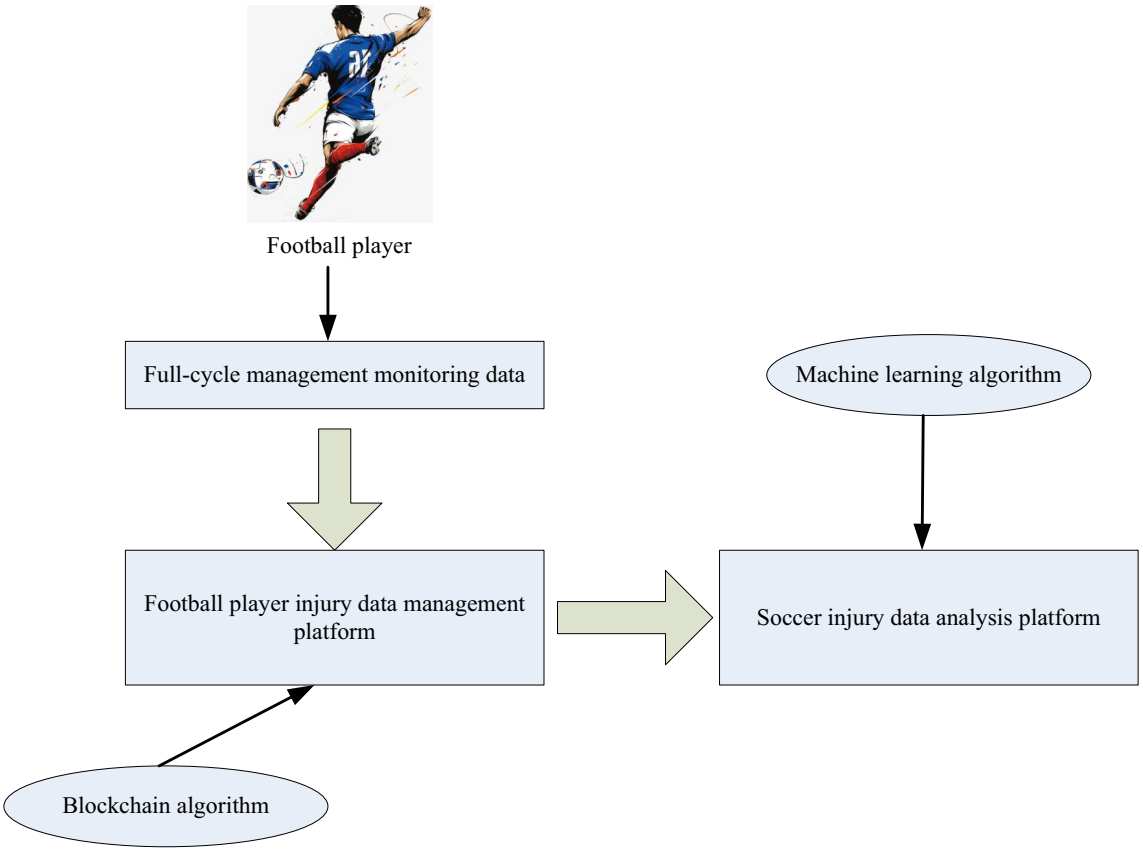


Fig. 4 Football player injury full-cycle management and supervision system model

4 Experiment of Full-Cycle Management and Supervision of Sports Injury

4.1 Data Source of Football Injury Management and Supervision

Sports events play a positive role in promoting social progress and economic development, but athletes are prone to sports injuries when playing sports [26, 27]. Football injury is the main cause of football players' performance, and there are many inducing factors of football injury. Therefore, this paper counts the injury data of 500 football players, mainly analyzing the causes of football injuries and the number of cases of various injury causes. The inducements of football players' injury are shown in Table 1.

Table 1 describes the main inducing factors of injury of football players. There were nine kinds of subjective and

Table 1 Incentives for injuries of football players

Damage factor category	Damage inducing factors	Number of cases (cases)	Proportion (%)
Subjective factor	Insufficient warm-up	80	16
	Technical nonstandard	30	6
	Insufficient stretching after exercise	40	8
	Insufficient awareness of safety precautions	70	14
	Psychological pressure	60	12
	Not knowing enough about physical fitness	60	12
Objective factor	Site factors	40	8
	Weather factors	50	10
	Competition arrangement factors	70	14

Table 2 Questionnaire results of football injury management and supervision system

Serial number	Index	Number of people (persons)	Proportion (%)
1	Self-processing capability of system	120	24
2	System data security	85	17
3	System operation convenience	90	18
4	System data carrying capacity	95	19
5	Comprehensiveness of system data	110	22

Table 3 Results of correlation analysis

Serial number	Index	Correlation (%)
1	Self-processing capability of system	28
2	System data security	22
3	System operation convenience	4
4	System data carrying capacity	26
5	Comprehensiveness of system data	20

objective factors. Insufficient warm-up was the most common injury in football, accounting for 16%. The proportion of football sports injuries caused by technical irregularities was at least 6%.

Football injuries do great harm to players. The whole cycle management and supervision system of football players' injuries was constructed, and the causes of injuries, treatment methods and subsequent recovery of football players were managed and monitored, so as to analyze effective measures for prevention and treatment of football players' injuries. To analyze the effect of football players' injury full-cycle management and monitoring system, this paper conducted a questionnaire survey on 200 professional football players, 100 football coaches and 200 football medical staff, which mainly investigated and evaluated the indicators of the effect of football players' injury full-cycle management and monitoring system. Table 2 shows the questionnaire survey results of football players' injury full-cycle management and monitoring system.

In Table 2, there were five indicators for evaluating the football injury management and monitoring system. The highest percentage of indicators was 24%, and the lowest percentage of indicators was 17%.

Due to the subjective differences in the results of the questionnaire survey, it is necessary to conduct a correlation analysis on the indicators evaluating the football injury management and supervision system. The results of the correlation analysis are shown in Table 3.

Table 3 describes the correlation of indicators for evaluating football injury management and monitoring system. The highest correlation of the system's self-processing capacity was 28%, followed by the system's data carrying capacity index, with a correlation of 26%. The minimum correlation of system operation convenience index was 4%. Since the system operation convenience index is less relevant than other indexes, the system operation convenience index would not be analyzed in the subsequent comparative experiment of football player injury full-cycle management and monitoring system.

4.2 Experimental Design of Full-Cycle Management and Supervision System

The football player injury full-cycle management system needs to store and analyze the football player injury full-cycle data. However, because the traditional management and supervision system mainly focuses on human resources, it is not only slow to collect football player injury data but also incomplete to analyze the data. The blockchain and machine learning algorithm were applied to the football player injury full-cycle MMS to intelligently process the football player injury data.

This paper draws on the latest research results [28], compared the football player injury full-cycle MMS based on blockchain and ML algorithm with the traditional football player injury full-cycle MMS by setting up a control group. The comparison points are system data security, system data comprehensiveness, system data carrying capacity and system self-processing capacity. To make the comparison between the two types of football players' injury full-cycle management and monitoring systems more sufficient, the experiment was set up for 5 months of system data testing. The performance of the two types of football player injury full-cycle management and monitoring systems was counted every other month. In addition, the injury data of football players tested in this experiment should also be different to make the comparison process closer to the real scene. The laboratory randomly selected 500 professional football players and 500 amateur football players as the objects for comparison of football players' injury full-cycle management and monitoring system.

5 Results of Sports Injury Full-Cycle Management and Supervision

5.1 System Data Security

In the football player injury full-cycle management and monitoring system, data acquisition and storage are jointly

completed by multiple servers. Although this method improves the diversity of injury data of football players, it also brings unsafe factors to the management and monitoring system. The tampering and omission of information would hinder the effect of sports injury full-cycle management and monitoring. The system data security of two types of football player injury full-cycle management and monitoring systems was compared, and the comparison results are shown in Fig. 5.

Figure 5a describes the comparison of the safety of two professional football players' injury full-cycle management and monitoring systems. Among them, the data security of the traditional football player injury full-cycle MMS reached a maximum of 58% in the third month and a minimum of 46% in the first month. However, the data security of the football player's injury full-cycle MMS based on blockchain and ML algorithm was constantly improving, and the system data security had increased from 56% in the first month to 82% in the fifth month. Figure 5b describes the comparison of the safety of two amateur football players' injury full-cycle MMS. Among them, the data security of the traditional football player injury full-cycle MMS was declining. This may be due to the excessive injury data of amateur athletes obtained by the management and supervision system during the experimental period, which led to the loss of many data and reduced the security of the system data. It decreased from 56% in the first month to 40% in the fifth month. However, the data security of the football player injury full-cycle MMS based on blockchain and ML algorithm was significantly higher than that of the traditional management and monitoring system, and the system data security reached 78% in the fifth month. Therefore, the application of blockchain and machine learning algorithm to the football player injury full-cycle management and monitoring system can significantly improve the security of system data.

5.2 Comprehensiveness of System Data

The comprehensiveness of the data in the football player injury full-cycle management and monitoring system would affect the judgment of football player injury. For example, it is necessary to record the specific causes of injury of football players, such as insufficient warm-up or nonstandard technical actions. Two kinds of football players' injury full-cycle management and monitoring systems were used to test the mobilization of professional football players and amateur football players, respectively. The data comprehensiveness comparison results of the two football player injury full-cycle management and monitoring systems are shown in Fig. 6.

Figure 6a describes the comparison of data comprehensiveness between the two professional football players' injury full-cycle management and monitoring systems. Among them, the comprehensiveness of the data of the traditional football player injury full-cycle management and monitoring system reached the highest 70% in the third month and the lowest 58% in the fifth month; the comprehensiveness of the average data was 64%. The data comprehensiveness of the football player injury full-cycle MMS based on blockchain and ML algorithm was constantly improving. It reached a minimum of 66% in the first month and a maximum of 86% in the fifth month, with the average data comprehensiveness of 76.8%. Figure 6b describes the comparison of data comprehensiveness between two amateur football players' injury full-cycle management and monitoring systems. Among them, the comprehensiveness of the data of the traditional football player injury full-cycle MMS had increased from 56% in the first month to 68% in the fifth month. The data comprehensiveness of the football

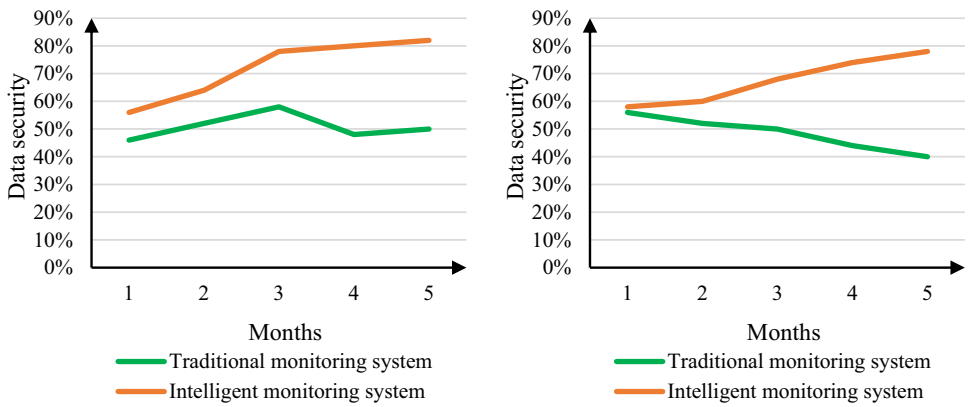


Fig. 5 Comparison results of system data security

(a) Safety of professional players' injury management and supervision system

(b) Safety of amateur players' injury management and supervision system

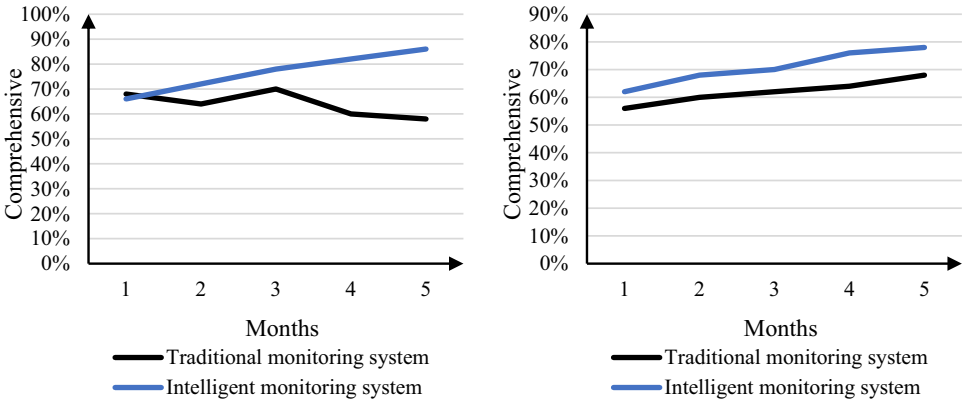


Fig. 6 Results of system data comprehensiveness comparison

(a) Systemic data comprehensiveness of professional player injuries

(b) Systemic data comprehensiveness of amateur player injuries

player injury full-cycle MMS based on blockchain and ML algorithm had increased from 62% in the first month to 78% in the fifth month. Therefore, the data obtained by the football player's injury full-cycle MMS using the blockchain and machine learning algorithm are more comprehensive.

5.3 System Data Carrying Capacity

The data storage methods of the traditional football player injury full-cycle management and supervision system and the football player injury full-cycle MMS based on blockchain and ML algorithm are different, which leads to different data carrying capacities of the two management and supervision systems. The former uses simple relational databases for data storage, while the latter uses blockchain for data storage. This paper compared the data carrying capacity of two kinds of football players' injury full-cycle management and monitoring systems. The comparison results are shown in Fig. 7.

Figure 7a describes the comparison of data carrying capacity of two professional football players' injury full-cycle management and monitoring systems. Among them, the data carrying capacity of the traditional football player injury full-cycle management and monitoring system reached the lowest of 44% in the fifth month and the highest of 52% in the third month. However, the data carrying capacity of the football player's full-cycle injury MMS based on blockchain and ML algorithm reached a maximum of 72% in the second month and a minimum of 66% in the third month. Figure 7(b) describes the comparison of the data carrying capacity of two amateur football players' injury full-cycle management and monitoring systems. The data carrying capacity of the traditional football players' injury full-cycle MMS reached the lowest 48% in the first month and the highest 60% in the fourth month. However, the data carrying capacity of the football player's injury full-cycle MMS based on blockchain and ML algorithm was constantly improving, from 66% in the first month to 80% in the fifth

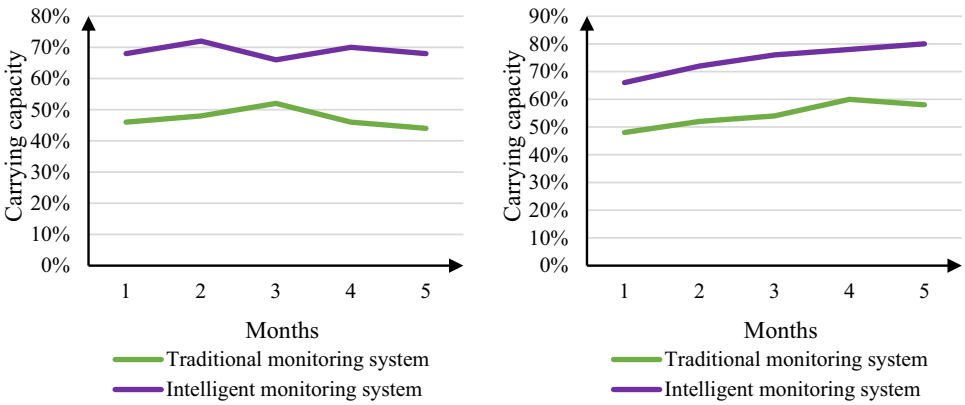


Fig. 7 Comparison results of system data carrying capacity

(a) System data carrying capacity of professional players' injuries

(b) System data carrying capacity of amateur players' injuries

month. Therefore, the football player injury full-cycle management and monitoring system with the blockchain as the data storage has a higher data carrying capacity.

5.4 Self-Processing Capacity of the System

The purpose of the full-cycle MMS for football player injuries is to record data on football player injuries. In addition, the cause of the injury is analyzed based on previous data, and the best injury treatment plan is provided. Therefore, the self-processing ability of football players' injury full-cycle management and supervision system is very important. The self-processing ability of the system is to measure the intelligence of the football player's injury full-cycle management and monitoring system. The comparison results of the self-processing ability of the two football player injury full-cycle management and monitoring systems are shown in Fig. 8.

Figure 8 describes the comparison results of the self-processing power of the two full-cycle injury management and monitoring systems of football players. Among them, the self-processing power of the traditional football player injury full-cycle management and monitoring system reached a minimum of 46% in the first month and a maximum of 54% in the third month, with an average self-processing power of 50%. The self-processing power of the football player injury full-cycle MMS based on blockchain and ML algorithms reached the highest level of 74% in the fifth month and the lowest level of 66% in the first month, with an average self-processing power of 70%. Therefore, the full-cycle MMS of football player injuries based on blockchain and ML algorithms can effectively improve the self-processing power of the system, and can make the management and monitoring system more intelligent.

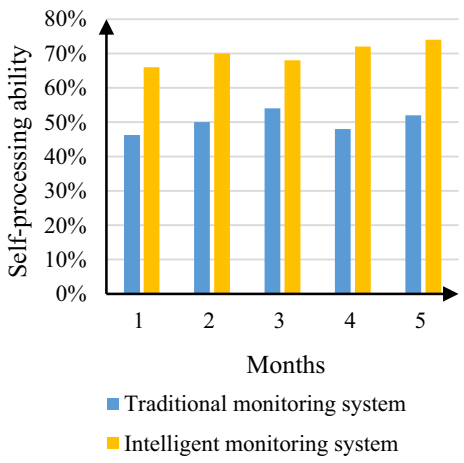


Fig. 8 Self-processing power comparison result diagram

6 Results and Discussion

Figure 5 shows the comparison results of system data security. The data security of traditional football player injury full-cycle MMS reached 58% in the third month, and the system data security increased from 56% in the first month to 82% in the fifth month. Figure 6 shows that the comprehensiveness of the full-cycle MMS data of injuries of traditional football players has increased from 56% in the first month to 68% in the fifth month. Figure 7 shows that the data carrying capacity of the full-cycle MMS for soccer players' injuries based on blockchain and ML algorithm has continuously improved, from 66% in the first month to 80% in the fifth month. These experimental data prove that compared with the traditional football player injury life cycle management and monitoring system, the football player injury life cycle MMS based on blockchain and ML algorithm has better data security, data integrity and data carrying capacity.

7 Conclusions

In recent years, soccer injuries have occurred frequently. It is necessary to systematically analyze the injury status, causes and how to prevent injuries of football players. However, the traditional football player injury full-cycle MMS has many defects. This paper applied the blockchain and machine learning algorithm to the football player injury full-cycle MMS, so that the football player injury full-cycle MMS has a complete sports injury information collection module, information storage module and information intelligent analysis module. Compared with the traditional football player injury lifecycle management and monitoring system, the football player injury lifecycle MMS based on blockchain and ML algorithm has better data security, data comprehensiveness and data carrying capacity. The football player's injury full-cycle MMS based on blockchain and ML algorithm can intelligently analyze and generate injury treatment schemes according to the injury situation of football players. However, in the process of comparing the two types of football player injury full-cycle management and monitoring systems, this paper only analyzed the data of 500 professional football players and 500 amateur football players, and the selected experimental objects were too few. Therefore, in the future, it would be the direction of future research to expand the comparison object of the two football players' injury full-cycle management and monitoring systems.

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Declarations

Conflict of Interest We confirm that there is no conflict of interest.

Ethical Approval This article does not contain any studies with human participants or animals performed by any of the authors.

Consent for Publication Written informed consent for publication was obtained from all participants.

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